
THE UNCONVENTIONAL WEEKLY

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UNCONVENTIONAL NEUROSCIENTISTS



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ABBVIE BETS \$1.2B ON PSYCHEDELICS TO TREAT DEPRESSION IN JUST 1 DOSE

Breakthrough alert: AbbVie invests \$1.2B in brexisillocin, a next-generation psychedelic therapy for depression, marking a leap where a few guided sessions could deliver months-long relief, signaling a bold shift in psychiatric treatment from chronic daily medication to rapid, neuroplasticity-driven interventions.

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What's Happening?

AbbVie's \$1.2 billion acquisition of brexisiloin from Gilgamesh Pharmaceuticals represents the largest Big Pharma investment in psychedelic medicine to date. This deal launches AbbVie into a \$50 billion psychedelic therapy market, signaling a shift from millions invested by biotech startups to billions from pharmaceutical giants. The move validates a treatment model that could replace daily antidepressants with 2-3 guided sessions producing benefits lasting six months or longer.

Brexisiloin also offers practical advantages over existing psychedelics. Whereas psilocybin or LSD can last 6-8 hours, brexisiloin works within a two-hour therapeutic window, making clinical administration more efficient while preserving extended benefits. In Phase 2 trials, a single 10 mg dose produced a 21.6-point reduction in depression scores by day 14, with 94% of patients achieving remission. Some participants reached near-zero depression scores by day 29, suggesting brexisiloin may outperform psilocybin.

The broader competitive landscape includes Johnson & Johnson's Spravato (ketamine-derived), psilocybin programs for depression and PTSD, synthetic LSD compounds, and MDMA. Regulatory challenges remain, highlighted by the FDA's recent rejection of MDMA for PTSD, but AbbVie's high-profile investment positions brexisiloin as a leading candidate in the race to define psychedelic medicine at scale.

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Why It Matters?

The "few doses, long-lasting cure" phenomenon represents a fundamental mechanistic breakthrough in psychiatry. Psychedelics like LSD and psilocybin bind directly to TrkB (BDNF receptor) with affinities 1,000-fold higher than traditional antidepressants, triggering neuroplasticity pathways that conventional treatments never accessed. Their potency also comes from penetrating neurons and activating intracellular 5-HT_{2A} receptors. Once ~70% of these receptors are activated, they trigger neuroplasticity cascades that persist long after the drug clears the system, rewiring neural circuits via TrkB, mTOR, and 5-HT_{2A} pathways after a single treatment.

Bretisilocin refines this mechanism further. Its optimized binding as a potent 5-HT_{2A} and 5-HT_{2C} agonist may drive more targeted activation of these pathways. Psilocybin studies have shown Default Mode Network disruption and enhanced brain integration and long-term remission, and bretisilocin's Phase 2 outcomes may amplify these effects, explaining its 94% remission rate.

Beyond efficacy, bretisilocin offers practical advantages. While psilocybin and LSD require 6–11 hours of clinical monitoring, bretisilocin delivers comparable therapeutic neuroplasticity in a two-hour window, reducing costs, improving accessibility, and enabling large-scale implementation. These combined benefits explain AbbVie's \$1.2 billion investment, validating psychedelic neuroplasticity as a platform technology with bretisilocin as the first compound optimized for mass clinical deployment.

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*Why It's **Unconventional**?*

Big Pharma is suddenly betting billions on compounds that were Schedule I just years ago, driven by clinical results surpassing decades of regulatory formalities. Neuroscience-driven drug discovery is happening in reverse: remarkable clinical outcomes drive investment first, with scientists pushing compounds into late-stage development second.

Most unconventionally, psychedelics show that depression isn't just a chemical imbalance, but a neural network dysfunction that can be “cured” through targeted neural circuit rewiring. Instead of daily pills, treatment involves periodic activation of the brain's natural plasticity in guided clinical sessions. The medicine isn't just the molecule but the therapeutic experience, blending neuropharmacology with psychology in ways that challenge traditional categories. This shift suggests the brain's capacity for self-reorganization is far more powerful than traditional neuroscience imagined, potentially revolutionizing how we treat every neuropsychiatric condition.

What's your **Unconventional** Take?

-  Is AbbVie's move into psychedelics a sign that mainstream pharma is ready to embrace therapies that work outside the traditional “daily pill” model?
-  By prioritizing speed to late-stage development, do companies risk pushing compounds into the market before we truly understand how they work?

*Are we entering a new era of brain medicine,
or just a more profitable one?*
